POSTNIKOV SYSTEM FROM THE SIMPLICIAL RESOLUTION IN THE MODULE CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-simplicial-resolution-ringed; external tool Derived Categories key Postnikov example; the two functor properties (Postnikov preservation, hocolim commutation) are the transport mechanism.

Editorial note. The label lemma-modules-postnikov names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Ringed simplicial resolution. Same unit, Lemma lemma-simplicial-resolution-ringed. For K = O the complex ... -> g_{2!}O_2 -> g_{1!}O_1 -> g_{0!}O_0 represents K = O; gives the special case.
- Key Postnikov example. Derived Categories, Example derived-example-key-postnikov. Constructs the Postnikov system of an object from a resolution; produces the triangles and the hocolim identification in the special case.

Source reference. Original definition 0D7Z: sources/stacks-chow/tags.txt, source lines 16574–16574. Prerequisite pointer invoked at proof line(s) 3467; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0D8Z: sources/stacks-chow/tags.txt, source lines 16609–16609. Prerequisite pointer invoked at proof line(s) 3475; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WI: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1711–1711. Prerequisite pointer invoked at proof line(s) 3461; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0D9B: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2092–2092. Prerequisite pointer invoked at proof line(s) 3485; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 3449; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$. Let K be an object of $D(\mathcal{C}_{total})$. Set*

$$X_n = (g_{n!}\mathcal{O}_n) \otimes^{\mathbf{L}}_{\mathcal{O}} K \quad \text{and} \quad Y_n = (g_{n!}\mathcal{O}_n \to \ldots \to g_{0!}\mathcal{O}_0)[-n] \otimes^{\mathbf{L}}_{\mathcal{O}} K$$

as objects of $D(\mathcal{O})$ where the maps are as in Lemma 09WI. With the evident canonical maps $Y_n \to X_n$ and $Y_0 \to Y_1[1] \to Y_2[2] \to \ldots$ we have

(1) *the distinguished triangles $Y_n \to X_n \to Y_{n-1} \to Y_n[1]$ define a Postnikov system (Derived Categories, Definition 0D7Z) for $\ldots \to X_2 \to X_1 \to X_0$,*
(2) *$K = hocolim Y_n[n]$ in $D(\mathcal{O})$.*

Proof. First, if $K = \mathcal{O}$, then this is the construction of Derived Categories, Example 0D8Z applied to the complex

$$\ldots \to g_{2!}\mathcal{O}_2 \to g_{1!}\mathcal{O}_1 \to g_{0!}\mathcal{O}_0$$

in $Ab(\mathcal{C}_{total})$ combined with the fact that this complex represents $K = \mathcal{O}$ in $D(\mathcal{C}_{total})$ by Lemma 0D9B. The general case follows from this, the fact that the exact functor $- \otimes^{\mathbf{L}}_{\mathcal{O}} K$ sends Postnikov systems to Postnikov systems, and that $- \otimes^{\mathbf{L}}_{\mathcal{O}} K$ commutes with homotopy colimits. □